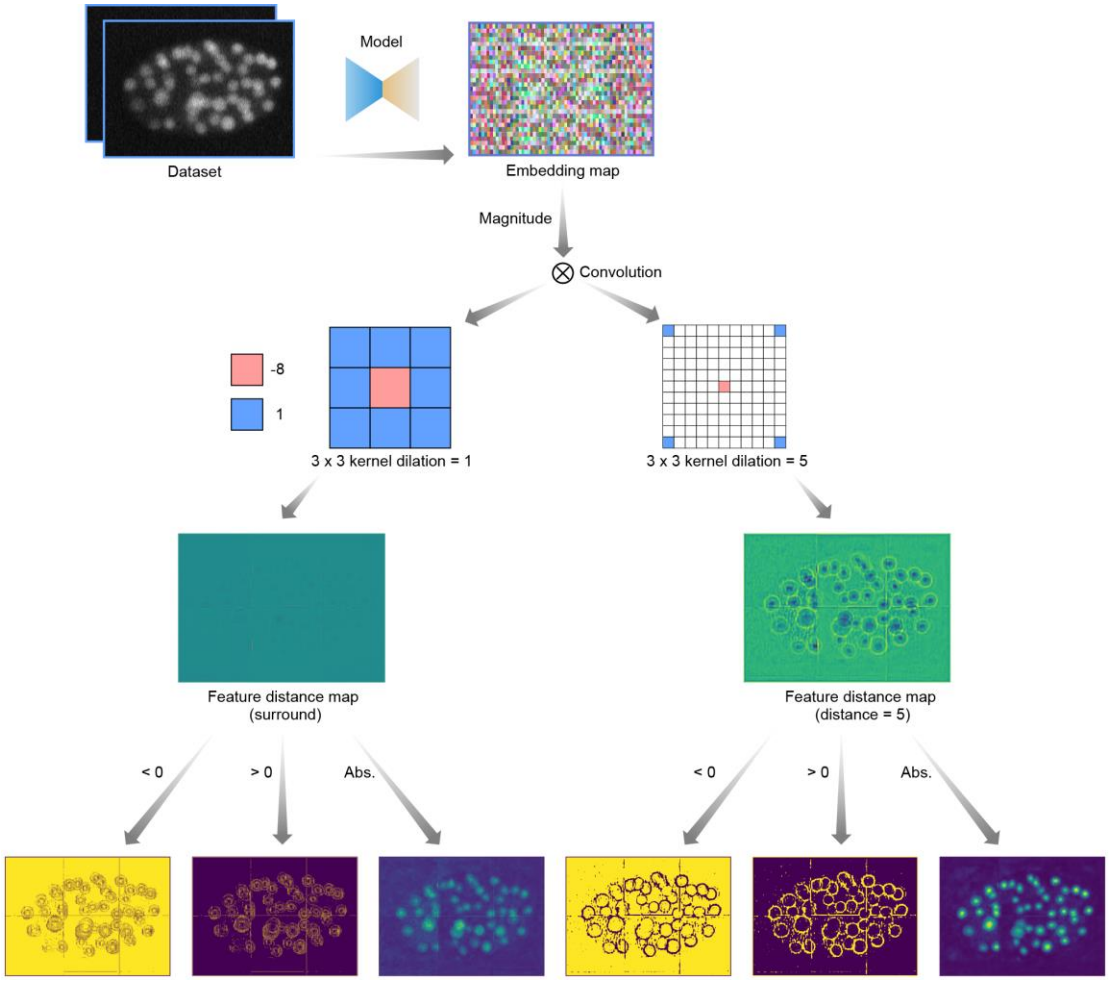


Supplementary Figure 4 | Visualization of the embedding map obtained by CELLECT. To analyze the spatial patterns captured by the model, we extracted embedding map corresponding to each voxel and computed their differences with surrounding voxels. This was achieved by convolving the squared values of the embedding map with a dilated convolution kernel, simulating neighborhood interaction. A dilation rate of 1 represents the average difference with adjacent voxels, while a dilation rate of 5 represents the difference with voxels at a distance of 5 voxels in all directions. The resulting feature distance map highlights the magnitude and positive or negative of feature differences between distinct cell regions in the image. At the boundaries, the opposing patterns between the central and surrounding areas particularly emphasize the differences in feature values between cellular and non-cellular regions. These differences can also assist in optimizing cell segmentation during the inference phase, especially under sparse annotation conditions. The artifacts observed at patch boundaries arise from the independent inference of each patch, which do not affect the accuracy of segmentation or tracking, as only the embedding vectors at predicted cell centers are used for downstream analysis, and the boundary variations are negligible. The visualization shown here is based on the Cell Tracking Challenge (CTC) Fluo-N3DH-CE dataset. The model used was trained on sequence #3 of the mskcc-confocal dataset.